

Reproducibility

Reproducibility I

This section explains how to regenerate every artifact in the study from a clean checkout. The exemplar's reproducibility guarantee is structural: each result is produced by a tested function and a thin script, never transcribed by hand.

How to regenerate everything I

From the repository root:

1. Run the analysis (writes figures + summary CSV, prints)

```
uv run python projects/templates/template_eda_notebook/src
```

2. Run the test suite with the coverage gate

```
uv run pytest projects/templates/template_eda_notebook/test
    --cov=projects/templates/template_eda_notebook/src --co
```

3. Render the manuscript

```
uv run python scripts/03_render_pdf.py --project templates,
```

Generated artifact registry I

The analysis script writes the following artifacts under `projects/templates/template_eda_notebook/output/`:

Artifact	Produced by
<code>figures/height_histogram.png</code>	<code>histogram_data()</code> + analysis script
<code>figures/correlation_heatmap.png</code>	<code>correlation_heatmap_data()</code> + analysis script
<code>figures/group_counts.png</code>	<code>group_count_data()</code> + analysis script
<code>data/summary_statistics.csv</code>	<code>summary_statistics()</code> + analysis script

Generated artifact registry II

The output/ tree is disposable and regenerated on every run; it is not the source of truth.

Determinism I

- ▶ The dataset (`data/measurements.csv`) is a static, committed fixture generated once with a fixed NumPy seed, so every statistic is reproducible bit-for-bit.
- ▶ The figure-data preparers are pure transforms with no RNG; the same inputs always produce the same bin counts, correlation values, and group counts.
- ▶ `clean_dataset()` reports exactly how many rows it removed, so the complete-case row count is a checkable invariant.

Verification (no hand-transcribed numbers) I

Every quantitative claim in [sec:results] is either reproduced by running the analysis script or registered in `data/claim_ledger.yaml` for evidence-registry validation. The manuscript intentionally does not embed volatile values, so prose and artifacts cannot disagree. The notebook itself is verified structurally by `tests/test_notebook.py`, which confirms it is valid nbformat, that its `from src` imports resolve to the library's public surface, and that no cell defines its own logic.